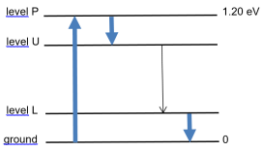


| Question |     |       | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Marks available |          |          |          |          |          |
|----------|-----|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------|----------|----------|----------|----------|
|          |     |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | AO1             | AO2      | AO3      | Total    | Maths    | Prac     |
| 7        | (a) | (i)   | More electrons occupy Level U than Level L<br>Accept more electrons in higher state than lower state.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 1               |          |          | 1        |          |          |
|          |     | (ii)  | All 3 arrows correct.<br>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                 | 1        |          | 1        |          |          |
|          |     | (iii) | Greater probability of stimulated emission [of photon] ... (1)<br>...than absorption [of photon] (1)<br>So extra <u>photon</u> produced (1)                                                                                                                                                                                                                                                                                                                                                                                                                                               | 3               |          |          | 3        |          |          |
|          | (b) |       | Calculation of $1.92 \times 10^{-19}$ J (1)<br>Use of $E = hf$ <b>or</b> $E = \frac{hc}{\lambda}$ (1)<br>$f < 2.9 \times 10^{14}$ Hz <b>ecf</b> or $\lambda > 1.0 \times 10^{-6}$ m <b>ecf</b> so visible light not produced (1)<br><br><b>Alternative:</b><br>Use of $\frac{hc}{e\lambda}$ (1)<br>eV values calculated at 400 nm and 700 nm i.e. 3.1 eV and 1.78 eV (1)<br>Lasing transition less than 1.20 eV so visible light not produced (1)<br><b>Alternative:</b><br>Lasing transition <1.20 eV (1)<br>Red photon is $\approx 2$ eV (1)<br>Therefore infra-red photon produced (1) |                 |          | 3        | 3        | 2        |          |
|          |     |       | <b>Question 7 total</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>4</b>        | <b>1</b> | <b>3</b> | <b>8</b> | <b>2</b> | <b>0</b> |